

Narrative Shift Detection in News Media Using Temporal Contrastive Learning

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github.com/meet6868/Narrative-Shift-Detection

Abstract

News media narratives evolve continuously in response to political, technological, and societal developments. Understanding these narrative shifts is crucial for media analysis, discourse tracking, and misinformation monitoring. Existing approaches—topic models, sentiment analysis, and document similarity methods—fail to explicitly model narrative shifts as temporal semantic phenomena with sentence-level interpretability.

We present a framework for narrative shift detection using Temporal Contrastive Learning (TCL) that tracks narrative evolution without supervised labels or topic-specific training. We collected and processed 1.17 million articles from 8 heterogeneous sources, applying rigorous cleaning (69.56% noise removal) to yield 355,334 English articles across five topics: War, Health, Technology, Climate, and Economics. Through sentence-level segmentation (3.08M sentences), context-aware SBERT embeddings (Window-5), and topic-aware filtering with curated prototypes from authoritative sources (WHO, Reuters, IPCC, IMF), we generated 1.34M topic-classified sentences.

Our TCL framework evolved through five experimental approaches, progressing from fixed windowing (InfoNCE loss) to adaptive segmentation using Ruptures-based change-point detection with multi-objective loss functions (NT-Xent + Topic Separation + Hard Negative Mining). Approach 4 achieves best inter-topic separation (-0.0875) and temporal consistency (0.9877) with 356 dynamic segments. We compare against two baselines: calibrated SBERT drift detection (4-stage pipeline, 15 topic-model threshold pairs) and K-Means clustering drift (K=5, Jensen-Shannon divergence).

1 Introduction

Narratives in news media are not static. As events unfold, political contexts change, and public sentiment shifts, the framing, emphasis, tone, and in-

terpretation of facts evolve over time. While multiple articles may discuss the same topic and entity, their narrative framing often changes gradually or abruptly. Understanding how these narratives change is essential for media behavior analysis, public discourse evolution, social science research, and misinformation monitoring.

Most existing approaches focus on topic change detection, sentiment trends, or document similarity. However, these approaches fail to explicitly model narrative shifts as a temporal semantic phenomenon that is entity-aware, topic-agnostic, learnable, and interpretable at the sentence level.

1.1 Motivation

News media narratives evolve continuously over time. Even when articles discuss the same topic, the framing, tone, emphasis, and intent often change in response to political, regulatory, technological, or societal developments. These changes, referred to as narrative shifts, play a crucial role in shaping public opinion and influencing policy discourse.

Existing approaches to analyzing narrative evolution face significant limitations. Topic models often conflate topic presence with narrative framing, while supervised change detection methods require labeled data that is scarce, subjective, and expensive to obtain. Recently, large language model (LLM)-based approaches have been explored, but they suffer from scalability, reproducibility, and interpretability concerns.

1.2 Objectives

This project proposes a generalized, scalable, and interpretable framework for detecting narrative shifts in news media without relying on labeled data, fixed topic definitions, or prompt-based LLM inference. Specifically:

1. Develop a generalized narrative shift detection

framework that works across multiple topics without topic-specific training

2. Implement temporal contrastive learning to model narrative evolution as a continuous temporal process
3. Provide sentence-level interpretability by identifying specific sentences that drive narrative changes
4. Build a large-scale multi-source news dataset across five major topics for comprehensive analysis
5. Design a scalable preprocessing pipeline that handles heterogeneous data sources and formats
6. Compare TCL against baseline approaches to validate effectiveness
7. Enable fine-grained temporal tracking of narrative evolution without requiring supervised labels

1.3 Problem Statement

Given a sequence of time-indexed news articles referring to the same real-world entity under a broad topic, automatically detect when narrative shifts occur, model how narratives change over time, identify specific sentences responsible for shifts, and explain how narrative framing has changed—all without requiring supervised narrative labels or topic-specific training.

Input: A topic T , entity E , and temporally ordered articles $A = \{(x_i, t_i)\}_{i=1}^n$ where x_i is article text and t_i is timestamp.

Output: Narrative shift scores, binary shift indicators, ranked sentences exhibiting strongest changes, interpretable explanations, time-ordered evolution trajectory, and narrative embeddings.

Task Framing: Framed as temporal contrastive learning where representations are learned such that stable narratives remain close in embedding space while shifted narratives are pushed apart.

2 Background and Related Work

2.1 Topic Modeling

Traditional topic modeling approaches such as Latent Dirichlet Allocation (LDA) and Dynamic Topic Models (Blei and Lafferty, 2006) discover latent topics in document collections and track their

evolution over time. However, these methods focus on topic distribution changes rather than semantic narrative shifts within topics, conflating topic presence with narrative framing.

2.2 Semantic Change Detection

Recent work in semantic change detection tracks how word meanings evolve using diachronic word embeddings (Hamilton et al., 2016). Methods like Word2Vec temporal alignment and BERT-based contextualized embeddings (Devlin et al., 2019) detect lexical semantic shifts. However, these approaches operate at word or phrase level and do not capture document-level or narrative-level semantic changes.

2.3 Document Similarity and Drift

Document similarity measures based on semantic embeddings (Reimers and Gurevych, 2019), cosine similarity, and clustering approaches detect concept drift in text streams. While useful for detecting distributional shifts, these methods lack temporal modeling capabilities and interpretability needed for narrative shift detection.

2.4 Contrastive Learning in NLP

Contrastive learning has emerged as a powerful paradigm for learning representations (Chen et al., 2020). In NLP, it has been successfully applied to sentence embeddings (Gao et al., 2021), document representations, and cross-lingual alignment. Temporal contrastive learning (Oord et al., 2018) extends this by incorporating temporal dynamics, learning representations where temporally adjacent samples with similar narratives are pulled together while narrative shifts push representations apart.

3 Data Details

3.1 Data Sources

We collected news articles from 8 heterogeneous sources spanning 2011–2025 (with extended coverage to 1970):

- **Kaggle Datasets (3 sources):** 1,096,988 articles (94.0%)
- **News APIs (3 sources):** 27,800 articles (2.4%)
- **Research Datasets (2 sources):** 42,259 articles (3.6%)

Total raw corpus: 1,167,047 articles.

3.2 Preprocessing Pipeline

Our preprocessing pipeline transforms raw articles into contextualized, topic-specific sentence embeddings through six stages:

Stage 0: Raw Collection — 1,167,047 articles from 8 sources

Stage 1: Deduplication — Removed 667,999 duplicate articles (57.24%), yielding 499,048 unique articles

Stage 2: Date Cleaning — Filtered articles with invalid timestamps, removing 38,459 articles (7.71%), yielding 460,589 articles

Stage 3: Language Filtering — Applied multi-threaded language detection, removing 105,255 non-English articles (22.86%), yielding 355,334 English articles. Processing time: 7.85 hours

Stage 4: Sentence Segmentation — Applied NLTK tokenization, generating 3,080,512 sentences (8.7 sentences/article) with context windows (W5: sentence ± 2 neighbors)

Stage 5: Context-Aware Embedding — Generated sentence embeddings using SBERT (all-mpnet-base-v2) with Window-5 context, producing $3,080,512 \times 768$ -dimensional embeddings

Stage 6: Topic Filtering — Applied cosine similarity (threshold 0.3) against curated topic prototypes from authoritative sources (WHO, Reuters, IPCC, IMF, MIT Tech Review), generating 1,335,158 topic-classified sentences (43.3% retention). Processing time: 28 minutes

3.3 Final Dataset Statistics

Table 1 shows topic-specific statistics after preprocessing.

Topic	Sentences	%	Size (GB)
War	490,123	36.7	4.59
Economics	277,886	20.8	2.60
Technology	190,543	14.3	1.78
Health	188,593	14.1	1.77
Climate	188,013	14.1	1.76
Total	1,335,158	100.0	12.50

Table 1: Topic-specific dataset statistics. All topics have 100% valid embeddings. Date range: 2011-09-06 to 2025-10-26.

3.4 Pipeline Statistics

Complete transformation from raw to processed data:

- **Raw $\rightarrow$ Clean (Article):** 30.44% retention (355,334 / 1,167,047)

- **Sentence Generation:** 8.7 sentences per article
- **Topic Filtering:** 43.3% sentence retention (1,335,158 / 3,080,512)
- **Processing Time:** 8.5 hours total
- **Output Size:** 12.50 GB across 5 topic files
- **Temporal Coverage:** 14+ years (2011-2025)
- **Quality Control:** 69.56% noise removal (811,713 articles removed)

3.5 Topic Prototype Construction

We manually curated 35 ideal reference articles (7 per topic) from authoritative sources:

- **War:** Reuters conflict coverage, BBC war reporting
- **Health:** WHO guidelines, medical journal articles
- **Technology:** MIT Technology Review, IEEE publications
- **Economics:** IMF reports, World Bank analyses
- **Climate:** IPCC reports, Nature Climate Change articles

Topic-defining sentences were manually extracted and embedded using SBERT to create 5 prototype vectors (768-dimensional), ensuring topic definitions are grounded in authoritative, domain-expert sources rather than arbitrary seed words.

3.6 Why Sentence-Level Analysis?

Traditional article-level approaches miss fine-grained narrative shifts. Consider:

“Artificial intelligence is rapidly expanding. Data centers now consume massive amounts of electricity. Governments are beginning to regulate AI usage.”

This single article touches on:

- **Technology** (AI expansion) — sentence 1
- **Climate** (energy consumption) — sentence 2
- **Policy/Economics** (regulation) — sentence 3

Sentence-level analysis allows us to identify the specific sentence discussing climate impact, track how climate framing of AI evolved over time, and detect when media narrative shifted from “AI innovation” to “AI energy concerns”—insights that article-level analysis would miss.

4 Proposed Methodology

4.1 Framework Overview

Our TCL framework combines sentence-level embedding, topic-aware filtering, temporal aggregation, and contrastive learning. The framework is: (1) Unsupervised—no labeled shift annotations, (2) Scalable—handles large corpora through efficient batching, (3) Interpretable—provides sentence-level explanations, (4) Topic-Aware—maintains context without topic-specific training, and (5) Temporally Continuous—models evolution as continuous process.

4.2 Temporal Contrastive Learning

Our TCL framework learns narrative representations by maximizing similarity between temporally adjacent sentences with stable narratives while pushing apart sentences exhibiting narrative shifts.

Encoder Architecture: Transformer-based encoder with 4 layers, 8 attention heads, feed-forward dimension of 1024, hidden dimension 512. Input features are 774-dimensional (768 SBERT + 5 topic scores + 1 normalized timestamp).

Multi-Objective Loss: We employ a weighted combination of three loss components:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{NT-Xent} + \lambda_2 \mathcal{L}_{topic} + \lambda_3 \mathcal{L}_{hard} \quad (1)$$

where $\mathcal{L}_{NT-Xent}$ is normalized temperature-scaled cross-entropy loss (Chen et al., 2020), $\mathcal{L}_{topic}$ encourages inter-topic separation, $\mathcal{L}_{hard}$ incorporates hard negative mining, and $\lambda_1 = 1.5, \lambda_2 = 0.5, \lambda_3 = 0.3$ are balancing coefficients determined through grid search.

NT-Xent Loss: For anchor embedding z_i , positive z_i^+ (temporally adjacent), and negatives $\{z_k\}_{k \neq i}$:

$$\mathcal{L}_{NT-Xent} = -\log \frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)} \quad (2)$$

where $\text{sim}(u, v) = u^T v / (\|u\| \|v\|)$ is cosine similarity and $\tau = 0.07$ is temperature.

Topic Separation Loss: Triplet-like loss (Schroff et al., 2015) ensuring minimum margin between topics:

$$\mathcal{L}_{topic} = \max(0, m - D_{inter} + D_{intra}) \quad (3)$$

where D_{inter} is inter-topic distance, D_{intra} is intra-topic distance, and margin $m = 0.5$.

Hard Negative Mining: Weights difficult negatives higher (Robinson et al., 2021):

$$\mathcal{L}_{hard} = -\log \frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\exp(\text{sim}(z_i, z_i^+)/\tau) + \sum_{z_h \in \mathcal{H}} \exp(\text{sim}(z_i, z_h)/\tau)} \quad (4)$$

where $\mathcal{H}$ are hard negatives (high similarity but different topic/time).

4.3 Adaptive Segmentation

Instead of fixed windowing, we employ Ruptures-based change-point detection (Truong et al., 2020) with PELT algorithm and RBF kernel. This enables adaptive segmentation identifying natural narrative boundaries.

The cost function for segment $[t_1, t_2]$ is:

$$\text{cost}(y_{t_1:t_2}) = \sum_{t=t_1}^{t_2} \|y_t - \mu_{t_1:t_2}\|_{RBF}^2 \quad (5)$$

where $\mu_{t_1:t_2}$ is mean embedding in segment. PELT finds optimal segmentation minimizing total cost with penalty $\rho = 0.1$ controlling segment count.

Maximum Similarity Grouping: After initial segmentation, we merge consecutive days when semantic similarity exceeds threshold $\theta_{merge} = 0.8$, stopping when similarity plateaus or decreases. This produces variable-length segments respecting narrative cohesion.

4.4 Approach Evolution

Our framework evolved through five experimental approaches, each addressing limitations of predecessors:

Approach 1: Fixed Window with InfoNCE

- Architecture: Transformer (4 layers, 8 heads, 256-dim)
- Windowing: Fixed 3-day windows (overlapping W=3, S=1 and non-overlapping W=3, S=3)

- Loss: InfoNCE contrastive loss
- Training: Layer normalization, gradient clipping, cosine annealing, warmup
- **Result:** Both intertopic and intratopic drift metrics poor
- **Samples Generated:** Varied by window strategy
- **Lesson:** Fixed windows insufficient for capturing narrative evolution

Approach 2: Grouping-Based Segmentation

- Architecture: Enhanced Transformer with attention pooling
- Grouping: Two strategies—(a) Count-based (4 days per group, 669 samples), (b) Date-range (5-day windows, 732 samples)
- Loss: NT-Xent + hard negative mining
- Feature: Topic-weighted lambda mean pooling
- **Result:** Intertopic drift very high (bad), intratopic drift improved
- **Issue:** Poor inter-topic separation despite grouping improvements
- **Lesson:** Grouping helps but requires better strategy and explicit topic modeling

Approach 3: Topic-Wise Window Sizing (Research Only)

- **Concept:** Different topics require different window sizes—War: 5 days, Health: 7 days, Climate: 8 days
- **Challenge 1:** Window size needs temporal adaptation *within* topics (e.g., Health: 3 days during COVID, 6-7 days otherwise)
- **Challenge 2:** Misses macro-level narrative shifts spanning multiple topics
- **Challenge 3:** Computationally expensive (5 topics $\times$ years $\times$ window sizes)
- **Status:** Not implemented; future work with RL/meta-learning

Approach 4: Dynamic Segmentation (Current Best)

- Architecture: Temporal Transformer (4 layers, 8 heads, 512-dim hidden)
- Segmentation: Ruptures PELT with RBF kernel (penalty=0.1)
- Features: 774-dim (768 SBERT + 5 topic scores + 1 time gap)
- Loss: Weighted NT-Xent ($\lambda = 1.5$) + Topic-Sep ($\lambda = 0.5$) + Hard-Neg ($\lambda = 0.3$)
- Grouping: Maximum similarity-based merging
- **Result:** Best inter-topic separation (-0.0875), temporal consistency (0.9877), 356 dynamic segments
- **Issue:** Entity-level confusion (e.g., “COVID vaccine” vs “flu vaccine”)
- **Lesson:** Need entity-aware modeling for fine-grained detection

Approach 5: Entity-Aware with NER (Planned)

- Architecture: Entity-aware Transformer with NER integration
- Features: 832+ dim (768 SBERT + 64 topic encoding + 1 time + entity features)
- Entity NER: SpaCy/BLINK/LUKE for PERSON, ORG, GPE, EVENT extraction
- Entity Embeddings: Entity-aware BERT representations
- Entity Attention: Multi-head attention with entity-query heads
- **Expected:** Resolve entity-level confusion, improve intratopic detection
- **Status:** Design complete; implementation future work

Table 2 summarizes the evolution of our TCL approaches.

Approach	Key Innovation	Loss Function	Inter-Topic	Intra-Topic	Status
1 (No Overlap)	Fixed window W=3, S=3	InfoNCE	-0.0457	0.2182	Completed
1 (Overlap)	Fixed window W=3, S=1	InfoNCE	-0.0286	0.1431	Completed
2 (Count)	4-day grouping	NT-Xent + Hard	0.0114	0.3365	Completed
2 (Date)	5-day windows	NT-Xent + Hard	-0.0361	0.3185	Completed
3	Topic-wise windows	N/A	N/A	N/A	Research
4	Ruptures segmentation	Multi-objective	-0.0875	0.9997	Current
5	NER + Entity-aware	Multi + Entity	TBD	TBD	Planned

Table 2: Evolution of TCL approaches. Approach 4 achieves best inter-topic separation (negative = good) and temporal consistency. Note: Approach 2 (Count) shows positive inter-topic score indicating poor separation.

4.5 Baseline Comparison Methods

All models (TCL, Semantic Drift, K-Means) use identical preprocessing: 1.33M sentences with 768-dim SBERT embeddings and topic scores, ensuring fair comparison.

Baseline 1: Semantic Drift Detection using SBERT

Four-stage calibration pipeline:

Stage 1—Topic Prototypes: Build canonical 768-dim vectors per topic from 35 hand-curated ideal articles (WHO, Reuters, IPCC, IMF). Mean-pool sentence embeddings, L2-normalize.

Stage 2—Relevance Thresholds: Determine minimum cosine similarity for on-topic classification. Formula: $\theta_{topic} = \mu_{sim} - 0.5\sigma_{sim}$. Calibrated thresholds: Climate 0.437, Economics 0.474, Health 0.441, Technology 0.437, War 0.412.

Stage 3—Drift Thresholds: Define shift threshold per topic/model. Process: Filter by relevance, group into 5-day windows, compute drift $d = 1 - \cos(w_t, w_{t-1})$. Formula: $\theta_{drift} = \mu_{drift} + \sigma_{drift}$. Example thresholds for W5: Climate 0.395, Economics 0.419, Health 0.439, Technology 0.409, War 0.480.

Stage 4—Detection: (1) Segment sentences, (2) Build context windows (W1/W3/W5), (3) Encode with SBERT, (4) Filter by topic, (5) Group into 5-day windows, (6) Compute drift, (7) Flag when $d > \theta_{drift}$, (8) Identify responsible sentences via impact score:

$$\text{impact}(s) = \frac{\cos(s, w_{curr})}{\cos(s, w_{prev}) + \epsilon} \quad (6)$$

Higher impact indicates sentence close to new narrative but far from old.

Characteristics: Multi-stage calibration, topic-specific thresholds, three context models (W1/W3/W5), interpretable output with responsible sentences, no temporal modeling between windows.

Baseline 2: K-Means Clustering Drift

Process: (1) Train K-Means with K=5 clusters on 768-dim SBERT embeddings, (2) Assign each sentence to nearest cluster, (3) Compute cluster distribution per date, (4) Calculate Jensen-Shannon divergence between consecutive dates: $JS(P\|Q) = \frac{1}{2}KL(P\|M) + \frac{1}{2}KL(Q\|M)$ where $M = (P + Q)/2$, (5) Flag shift when $JS > 0.3$.

Characteristics: Captures distributional changes, unsupervised, K selection critical, no explicit temporal modeling, sensitive to cluster number, struggles with gradual shifts, interpretable through cluster representatives.

5 Experimental Results

5.1 Data Pipeline Completion

Successfully completed comprehensive data processing:

- **Integration:** 8 heterogeneous sources with robust CSV/Excel/JSON parsing
- **Language Detection:** Multi-threaded implementation achieving $8\times$ speedup (16 articles/second)
- **Cleaning:** Multi-stage pipeline with aggressive 69.56% noise removal
- **Segmentation:** 3.08M sentences with Window-5 context preservation
- **Embedding:** GPU-accelerated batch processing for $3.08M \times 768$ -dim embeddings
- **Topic Filtering:** 1.34M topic-classified sentences across 5 topics (12.5 GB)
- **Memory Management:** Incremental processing preventing overflow
- **Quality Metrics:** 99.8% data completeness, 100% valid embeddings

5.2 TCL Training Results

Table 3 presents comprehensive performance metrics across all TCL approaches.

Key Metrics Interpretation:

Inter-Topic Drift (Goal: Low/Negative): Measures topic distinctness in embedding space. Approach 4 achieves -0.0875, indicating excellent separation where War articles are clearly distinct from Health articles. Approach 2 (Count-based) shows positive 0.0114, indicating topics overlap (model confuses War with Health).

Intra-Topic Drift (Goal: Moderate): Measures similarity of consecutive windows within same topic. Approach 4's 0.9997 is very high, potentially indicating over-smoothing where all windows appear identical (may miss subtle shifts). Approach 2's 0.3185 is moderate, detecting genuine narrative shifts while maintaining stability.

Separation Score: Quantifies topic distinctiveness. All approaches currently below ideal threshold of 2.0, indicating need for refinement. Approach 2 (Count) shows extreme -29.56, indicating catastrophic topic overlap.

Temporal Consistency: Measures narrative segment cohesion. Approach 4's 0.9877 indicates highly cohesive segments with natural narrative boundaries.

Sample Count: Approach 4 produces 356 dynamic segments (fewer than fixed approaches' 669-732), indicating adaptive segmentation identifies longer, more meaningful narrative units.

5.3 Approach-Specific Analysis

Approach 1: Fixed Windowing

Tested two variants—non-overlapping (W=3, S=3) and overlapping (W=3, S=1):

Non-Overlapping: Produced discrete 3-day windows with no context sharing. Issues: (1) Sharp boundaries miss gradual transitions, (2) Uniform 3-day assumption inappropriate (War needs shorter, Climate needs longer), (3) Overlapping sentences lost between windows.

Overlapping: Sliding 3-day window (stride=1) maintains continuity. Issues: (1) Excessive similarity between adjacent windows (similarity 0.95+), (2) Model struggles to distinguish genuine shifts from temporal proximity, (3) Computational overhead for marginal benefit.

Performance: Both variants show weak inter-topic separation (-0.0457, -0.0286) and poor intra-topic scores (0.2182, 0.1431), indicating fixed win-

dowing fundamentally insufficient for narrative modeling.

Approach 2: Grouping Strategies

Tested count-based (4 consecutive days) and date-gap (merge dates within 3-day range):

Count-Based: 669 fixed-size groups. Issues: (1) Catastrophic inter-topic confusion (score +0.0114), (2) Extreme separation score (-29.56) indicates topics completely overlap, (3) Fixed grouping ignores semantic content—forces unrelated dates together.

Date-Gap: 732 variable groups based on temporal proximity. Improvements: (1) Better inter-topic separation (-0.0361), (2) Similar intra-topic performance (0.3185), (3) More natural grouping respecting temporal gaps.

Key Insight: Date-gap grouping outperforms count-based by respecting temporal structure, but both suffer from predetermined grouping strategies not grounded in semantic content. Need data-driven segmentation.

Approach 4: Dynamic Segmentation (Current Best)

Used Ruptures PELT algorithm with RBF kernel for adaptive segmentation:

Segmentation Process: (1) PELT identifies change-points in 768-dim embedding time series based on statistical properties, (2) Penalty parameter 0.1 controls granularity (lower = more segments), (3) Maximum similarity grouping merges semantically similar consecutive segments.

Results: 356 dynamic segments (fewer than fixed approaches) with excellent inter-topic separation (-0.0875) and temporal consistency (0.9877).

Strengths: (1) Data-driven boundaries align with actual semantic shifts, (2) Variable-length segments accommodate topic-specific pacing (War: short intense segments, Climate: long gradual segments), (3) Explicit topic separation loss improves multi-topic modeling.

Weaknesses: (1) Very high intra-topic similarity (0.9997) suggests over-smoothing, (2) Entity-level confusion observed—model correctly matches topics but fails to distinguish specific entities ("COVID vaccine" vs "flu vaccine" both tagged as Health shifts), (3) Separation score (1.09) below ideal threshold.

Entity Confusion Example: Tested on Health topic sentences:

- Sentence A (2020): "COVID-19 vaccines show promise in trials"

Approach Config		Samples	Inter-Topic	Intra-Topic	Separation	Temp. Consist.
1	W=3, S=3 (NoOverlap)	Variable	-0.0457	0.2182	-4.78	0.9155
1	W=3, S=1 (Overlap)	Variable	-0.0286	0.1431	-5.01	0.8978
2	Fixed 4-day groups	669	0.0114	0.3365	-29.56	0.9193
2	Max 3-day gap	732	-0.0361	0.3185	-8.83	0.8948
4	Dynamic Ruptures	356	-0.0875	0.9997	1.09	0.9877

Table 3: Detailed TCL performance metrics. Inter-Topic: lower (more negative) = better separation. Intra-Topic: moderate values ideal (too high = over-smoothing, too low = noise). Separation Score: ideal > 2.0. Temporal Consistency: higher = more cohesive segments.

- Sentence B (2015): “Annual flu vaccines recommended for elderly”

Model outputs: High topic similarity (both Health), High shift score (different time periods), but semantically these represent different entities. Approach 4 lacks entity-awareness to distinguish.

5.4 Baseline Comparison

Preliminary experiments comparing TCL Approach 4 against baselines across Climate topic show that TCL detects more narrative shifts with higher semantic distances at shift points, demonstrating superior sensitivity to narrative changes. K-Means clustering provides a fast unsupervised baseline but misses gradual shifts due to fixed cluster assumptions. SBERT drift detection with calibrated thresholds captures major distributional changes but requires manual threshold tuning for each topic.

Analysis:

K-Means Drift: Detects 18 shifts with score 0.31. Strengths: Fast, unsupervised, captures major distributional changes. Weaknesses: Fixed K=5 assumption, misses gradual shifts, sensitive to cluster initialization, no temporal modeling (treats dates independently).

SBERT Drift (W5): Detects 26 shifts with score 0.38. Strengths: Topic-aware filtering, calibrated thresholds, interpretable responsible sentences, three context models. Weaknesses: Threshold-dependent (conservative), fixed 5-day windows, no learning between windows, may miss subtle framing changes.

TCL Approach 4: Detects 32 shifts with score 0.44. Strengths: Highest sensitivity, adaptive segmentation, learned representations capture gradual evolution, explicit temporal modeling. Weaknesses: Slowest (245s vs 89s for SBERT), requires training, more complex pipeline.

Semantic Distance Validation: Higher average cosine distance at detected shift points (TCL: 0.56 ; SBERT: 0.48 ; K-Means: 0.41) suggests TCL

detects more semantically meaningful shifts rather than noise.

Shift Detection Examples: Manual inspection of top-10 shifts per model:

K-Means: Detected major events (Paris Agreement 2015, Trump withdrawal 2017), but missed gradual policy discussions and scientific consensus shifts.

SBERT: Captured both major events and some policy framing changes, identified when discussion shifted from “climate change science” to “climate action policy.”

TCL: Detected all major events plus subtle framing evolution—e.g., transition from “mitigation strategies” (2012-2014) to “adaptation measures” (2015-2017) to “climate justice” (2018-2020). Captured gradual narrative maturation often missed by threshold-based methods.

5.5 Quantitative Evaluation Metrics

We evaluate all models as drift detection systems using four unsupervised metrics:

1. Drift Score Distribution Analysis

This metric compares the statistical properties of shift scores across models. We compute the mean shift magnitude to measure average narrative change intensity, standard deviation to assess detection sensitivity (higher variance indicates better ability to distinguish genuine shifts from noise), and count high-confidence shift events with scores exceeding a threshold. Models with higher variance and more high-shift events demonstrate superior detection capabilities.

2. Drift Peak Detection

This approach counts detected shifts using multiple threshold strategies to assess robustness. We employ three methods: adaptive thresholding based on mean plus one standard deviation, percentile-based detection identifying the top fraction of shifts, and fixed threshold comparison. Consistent performance across diverse threshold strategies in-

icates model reliability and reduces sensitivity to hyperparameter selection.

3. Semantic Distance at Shift Points

To validate that detected shifts correspond to genuine semantic changes rather than spurious signals, we compute the average cosine distance between consecutive sentence embeddings at identified shift points. Higher distances indicate more meaningful narrative transitions, while lower distances suggest potential false positives where the model flags minor variations as significant shifts.

4. Human Validation (Primary Metric)

Human validation provides the most reliable assessment of narrative shift detection quality. We manually annotate the highest-confidence detected shifts per model, asking expert annotators to judge whether each represents a genuine narrative or topic change. Multiple independent annotators review sentences before and after each shift point to ensure reliability. Inter-annotator agreement is measured using Cohen’s Kappa statistic, with disagreements resolved through discussion. We analyze false positives to identify systematic model weaknesses, categorizing errors by root cause such as algorithmic artifacts, data quality issues, or fundamental modeling limitations.

5.6 Ablation Studies

Conducted ablation studies on Approach 4 to understand component contributions:

Component Analysis:

Removing Topic Separation Loss: Inter-topic score degrades from -0.0875 to -0.0423 (51% worse), confirming explicit topic modeling is critical. Without this loss, model conflates similar but distinct topic narratives.

Removing Hard Negative Mining: Moderate degradation in inter-topic (-0.0761) and larger impact on intra-topic (0.9945) and temporal consistency (0.9653). Hard negatives help distinguish difficult cases.

Removing Ruptures Segmentation (using fixed 5-day windows): Significant degradation especially in intra-topic (0.8234) and temporal consistency (0.8921). Demonstrates adaptive segmentation’s importance for natural narrative boundaries.

NT-Xent Only (no topic loss, no hard negatives): Baseline performance similar to Approach 1, confirming need for multi-objective optimization.

6 Project Timeline

Our project development followed a structured timeline from initial conception through final implementation:

Phase 1: Data Collection & Preprocessing (Weeks 1-3)

- Identified and integrated 8 heterogeneous news sources (Kaggle datasets, news APIs, research corpora)
- Implemented multi-stage cleaning pipeline: deduplication (57.24% removal), date validation (7.71% removal), language filtering (22.86% removal)
- Developed sentence segmentation with context windows (Window-1, Window-3, Window-5)
- Generated SBERT embeddings for 3.08M sentences using GPU acceleration
- Curated 35 authoritative reference articles for topic prototypes
- Applied topic filtering yielding 1.34M classified sentences across 5 topics

Phase 2: Baseline Implementation (Weeks 4-5)

- Implemented K-Means clustering drift detection with Jensen-Shannon divergence
- Developed 4-stage SBERT calibration pipeline with topic-specific thresholds
- Validated baselines on sample datasets, tuning hyperparameters

Phase 3: TCL Framework Development (Weeks 6-9)

- Approach 1: Fixed windowing strategies (overlapping and non-overlapping)
- Approach 2: Grouping-based segmentation (count-based and date-range)
- Approach 3: Research phase exploring topic-wise adaptive windows
- Approach 4: Ruptures-based dynamic segmentation with multi-objective loss
- Implemented NT-Xent, topic separation, and hard negative mining losses

Configuration	Inter-Topic	Intra-Topic	Temp. Consist.
Full Model	-0.0875	0.9997	0.9877
- Topic Sep Loss	-0.0423	0.9989	0.9802
- Hard Neg Mining	-0.0761	0.9945	0.9653
- Ruptures Seg	-0.0512	0.8234	0.8921
NT-Xent Only	-0.0389	0.7812	0.8567

Table 4: Ablation study results. Each row removes one component. Topic Separation Loss critical for inter-topic distinction. Ruptures Segmentation essential for temporal consistency.

- Trained transformer encoders (4 layers, 8 heads, 512-dim hidden state)

Phase 4: Evaluation & Analysis (Weeks 10-11)

- Cross-topic performance evaluation across all 5 topics
- Ablation studies quantifying component contributions
- Human validation experiments with inter-annotator agreement analysis

Phase 5: Documentation & Finalization (Week 12 - March 8, 2026)

- Comprehensive documentation of methodology and results
- Code refactoring and repository organization
- Final report preparation and submission

Future Work: Planned Approach 5 will incorporate entity-aware modeling using NER (SpaCy/BLINK) to resolve entity-level confusion, addressing current limitation of conflating different entities within same topic.

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